

Average

1. The average marks in English subject of a class of 24 students is 56. If the marks of three students were misread as 44, 45 and 61 against actual marks of 48, 59 and 67 respectively, then what would be the correct average?
(A) 56 (B) 55
(C) 57.5 (D) 58.5
(E) None of these
2. A student took 4 papers in an examination, where the maximum marks were the same for each paper. In all papers together, the candidate obtained 60% of the total marks. His marks in these papers were in the ratio of 24:30:32:34. Then in how many of the papers he got more than 60% marks?
(A) 6 (B) 5
(C) 4 (D) 3
(E) 2
3. The average weight of A, B, C and D is 33 kgs. C is 12kgs lighter than A while B is 6 kg heavier than C. D's weight is 10 kgs less than that of C. What is the average weight of A and D?
(A) 32 kg (B) 33 kg
(C) 34 kg (D) 35 kg
(E) None of these
4. In an office of 40 people, the average salary of class-A, class-B and class-C officer is Rs600, Rs750 and Rs. 1000 a day respectively and there ratio of strength is 5:4:1 respectively. Find the monthly average salary of an officer (in Rs.)? Assume the number of days in a month to be 30.
(A) 21000 (B) 24600
(C) 25800 (D) 24030
(E) None of these
5. The average age of 29 students in a class is 26 years. If the age of teacher is also included the average age of the class becomes 26.4 years. Find the age of the teacher.
(A) 32 years (B) 36 years
(C) 42 years (D) 38 years
(E) 48 years
6. There are 45 students in a hostel. If the number of student increases by 5, the expenses of the mess increases by Rs.30 per day while the average expenditure per head diminishes by Rs.2. Find the original expenditure of the mess.
(A) Rs. 1200 (B) Rs. 1170
(C) Rs. 1710 (D) Rs. 1250
(E) Rs. 1240
7. In a set of 6 numbers, the average of first four is 10 and that of last four is 8. If the average of all the six numbers is 9, then find the average of the third and fourth number.
(A) 7 (B) 8
(C) 9 (D) 10
(E) 11
8. Three Science classes A, B and C take a Life Science test. The average score of students of class A is 83. The average score of students class B is 76. The average score of class C is 85. The average score of class A and B is 79 and average score of class B and C is 81. Then the average score, of classes A, B and C is
(A) 80 (B) 80.5
(C) 81 (D) 81.5
(E) None of these

9. The second lowest number of five consecutive odd number series is four more than the $\frac{5}{12}$ th of the third highest number of a five consecutive even number series. If the average of five consecutive even number series is 60, then find the difference between the highest number of both the series?
(A) 27 (B) 29
(C) 31 (D) 33
(E) 37
10. In the given questions, two quantities are given, one as Quantity I and another as Quantity II. You have to determine relationship between two quantities and choose the appropriate option.
Quantity I: Three boys of a class of weight 75, 68 and 57 are replaced by 23, 41 and 56. Due to which average weight of class is decreased by 2.5. What is the strength of class.
Quantity II: 50
(A) Quantity I > Quantity II
(B) Quantity I < Quantity II
(C) Quantity I $\geq$ Quantity II
(D) Quantity I $\leq$ Quantity II
(E) Quantity I = Quantity II or no relation
11. The average marks of 100 students were found to be 40. Later on it was discovered that a score of 53 was misread as 83. Found the correct average corresponding to the correct score.
(A) 38.7 (B) 39
(C) 39.7 (D) 41
(E) None of these
12. The average age (in years) of a group of people is twice the number of people in the group. A person, X, leaves the group and the average age is still twice the number of people in the group. Now another person, Y, leaves the group and the average age is still twice the number of people in the group. If the ratio of the ages of X and Y is 19 : 17, then find the average age of the group, if one more person, Z, of age 16 years, leaves the group.
(A) 10 (B) 15
(C) 16 (D) 18
(E) None of these
13. A student studied four subjects in his bachelor's degree. If the ratio of marks in subject A and B is 1:2, in subject B and C is 2:5 and in subject A and D is 1:4. If she has scored an average of 84 marks in a subject. In how many subjects did she score equal to or greater than 60 marks?
(A) 1 (B) 2
(C) 3 (D) 4
(E) Cannot be determined
14. Average score of Rahul, Manish and Suresh is 63. Rahul's score is 15 less than Ajay and 10 more than Manish. If Ajay scored 30 marks more than the average score of Rahul, Manish and Suresh, what is the sum of Manish's and Suresh's scores?
(A) 120 (B) 111
(C) 117 (D) Cannot be determined
(E) None of these
15. 12 years ago, there were 7 members in Arun's family & then the average age of the family was 28 years. His father died three years ago from present & then the average age of the family was 26 years. Then his father died at the age of-
(A) 88 years (B) 93 years
(C) 103 years (D) 106 years
(E) 97 years
16. The average weight of three men A, B and C is 65 kg. Another man D joins the group and the average now becomes 68 kg. If another man E whose weight is 4 kg less than that of D, replaces A then the average weight of B,

C, D and E becomes 67 kg. Find the weight of A?

- (A) 75 kg (B) 78 kg
(C) 77 kg (D) 82 kg
(E) 86 kg

17. The average wages of four farmers in a field is Rs. 60. If the difference between the wages of the farmer with the lowest wages and second lowest wages, difference between the wages of the farmer with the second lowest and third lowest wages and so on is Rs. 20, then wage of a farmer with least wage is -
(A) Rs. 30 (B) Rs. 80
(C) Rs. 50 (D) Rs. 40
(E) Rs. 60

18. The average age of 40 teachers in a school in 2007 was 35 years. In 2009, 10 teachers retired from the job, whose average age was 50 years. In 2012, 20 teachers joined the school, whose average age was 30 years. What was the total age of all the teachers in 2015 assuming that no teacher joined or left the school apart from the ones mentioned in the question?
(A) 1450 years (B) 1560 years

- (C) 1820 years (D) 1720 years
(E) None of these

19. The average age of 100 bankers in PNB in 2002 was 50 years. In 2004, 20 bankers retired from their posts, whose average age was 60 years. After a certain gap in 2007, 40 bankers joined PNB whose average age was 38 years. The average age of all the bankers in 2010 was?
(A) 45 years (B) 51 years
(C) 60 years (D) 40 years
(E) 48 years

20. In a college, 180 students took admission in Computer science department, of which 100 are girls and rest are boys. The average marks of 80 boys and 100 girls was calculated as 124 and 92 respectively. However, it was later found that the marks of one of the girl were read as 122 while her actual marks were 82. Similarly in the case of boys, the marks of one boy were read as 110 instead of 130. What is the actual average marks of 180 students?
(A) 117.6 (B) 112.4
(C) 106.1 (D) 99.8
(E) 115.4

Average

Solution

1. **Answer: (E)**

Correct average = $(56 \times 24 - 44 - 45 - 61 + 48 + 59 + 67)/24 = 57$

2. **Answer: (E)**

Let the student scored $24x$, $30x$, $32x$ and $34x$ marks in 4 papers. Therefore, sum of marks in all the papers together = $12x + 15x + 16x + 17x = 60x$.

Average of the total marks obtained = 60%

Let the maximum marks in a paper be 100

Then, $60x = 60$ (4)

$x = 4$

The marks obtained in 4 papers are 48, 60, 64, 68. He got more than 60% of the marks in 2 papers.

3. **Answer: (A)**

Let the weight of A, B, C and D be a , b , c and d kg respectively.

$c = a - 12$

$b = c + 6$

$\Rightarrow b = a - 6$

$d = c - 10$

$\Rightarrow d = a - 22$

Given, $a + b + c + d = 132$

Putting the values of a , b , c and d ,

$a + (a - 6) + (a - 12) + (a - 22) = 132$

i.e. $a = 43$

And $d = a - 22 = 21$

Average of A and D is $(43 + 21)/2 = 32$.

4. **Answer: (A)**

Average salary = total salary/total strength

No. of class A officers = $5/10 \times 40 = 20$

No. of class B officers = $4/10 \times 40 = 16$

No. of class C officers = 4

Average salary = $(600 \times 20 + 750 \times 16 + 1000 \times 4)/40 = \text{Rs. } 700$

Average monthly salary

= $700 \times 30 = \text{Rs. } 21000$

5. **Answer: (D)**

Total age of students of the class = (29×26)

= 754 years

Total age of a class including teacher = $(30 \times 26.4) = 792$ years

So, Age of teacher = $(792 - 754) = 38$ years.

6. **Answer: (B)**

let the average expenditure per student per day be Rs. x .

then total expenditure per day = $45x$

new total expenditure per day

= Rs $(45x + 30)$

new average expenditure = $x - 2$

given, $(45x + 30)/50 = x - 2$

thus, $x = 26$

Now original expenditure of the mess

= $45 \times 26 = \text{Rs. } 1170$

Hence, option b.

7. **Answer: (C):**

Sum of first four numbers = $10 \times 4 = 40$

Sum of last four numbers = $8 \times 4 = 32$

Sum of all six numbers = $9 \times 6 = 54$

Now,

Sum of the 3rd and 4th numbers = Sum of first 4 numbers + Sum of last four numbers

– Sum of all six numbers = $40 + 32 - 54 = 18$

$\therefore$ Average of 3rd and 4th numbers = $18/2 = 9$

8. **Answer: (D)**

Let no. of student in class A, B and C be x , y and z

$\therefore A = 83x$

$B = 76y$

$C = 85z$

Now, $A + B = 79x + 79y$

$B + C = 81(y + z) = 81y + 81z$

$\therefore 83x + 76y = 79x + 79y$

$4x = 3y$

$\frac{x}{y} = \frac{3}{4}$

And, $76y + 85z = 81y + 81z$

$5y = 4z$

$\frac{y}{z} = \frac{4}{5}$

$\therefore x : y : z = 3 : 4 : 5$

$$\begin{aligned}\therefore \text{Required average} &= \frac{83 \times 3 + 76 \times 4 + 85 \times 5}{12} \\ &= \frac{249 + 304 + 425}{12} \\ &= \frac{978}{12} \\ &= 81.5\end{aligned}$$

9. **Answer: (B)**

Let five consecutive even number series
= $a, (a + 2), (a + 4), (a + 6), (a + 8)$

Five consecutive odd number series

ATQ –

$$\frac{a + (a + 2) + (a + 4) + (a + 6) + (a + 8)}{5} = 60$$

$$5a + 20 = 300$$

$$a = 56$$

Third highest number of even series

$$= 56 + 4 = 60$$

Second lowest number of odd number series

$$= 60 \times \frac{5}{13} + 4 = 29$$

Highest number of even number series

$$= (56 + 8) = 64$$

Highest number of odd number series

$$= (29 - 2 + 8) = 35$$

$$\text{Required difference} = 64 - 35 = 29$$

10. **Answer: (B)**

I. Weight of three student initially
= $75 + 68 + 57 = 200$

Weight of three students which are replaced

$$= 32 + 42 + 56 = 120$$

$$\text{Difference} = 200 - 120 = 80$$

This 80 is divided onto all students

$$\text{Total students} = \frac{80}{2.5} = 32$$

II. 50

Quantity I < Quantity II

11. **Answer: (C)**

Average of 100 student = 40

$$\text{Total} = 40 \times 100 = 4000$$

$$\text{Error} = 83 - 53 = 30 \text{ (high)}$$

$$\text{So correct average} \Rightarrow \frac{3970}{100} = 39.7$$

12. **Answer: (C)**

Let, initial no. of people in the group be 'n'.

Let $19x$ and $17x$ be ages of X and Y respectively,

ATQ,

$$2n^2 - 19x = 2(n - 1)^2 \dots\dots\dots(i)$$

$$\text{And } 2n^2 - 19x - 17x = 2(n - 2)^2$$

$$2n^2 - 36x = 2(n - 2)^2$$

$\dots\dots\dots(ii)$

Solving (i) and (ii),

$$x = 2, n = 10$$

Average age of group after Z leaves the group

$$= \frac{2 \times 10^2 - 19 \times 2 - 17 \times 2 - 16}{10 - 3} = \frac{112}{7} = 16$$

13. **Answer: (B)**

$$A : B = 1 : 2$$

$$B : C = 2 : 5$$

$$A : D = 1 : 4$$

$$\text{Thus, } A : B : C : D = 1 : 2 : 5 : 4$$

$$(x + 2x + 5x + 4x) / 4 = 12x / 4 = 3x = 84$$

$$\text{So, } x = 28$$

Marks in subject A = 28

Marks in subject B = 56

Marks in subject C = 140

Marks in subject D = 112

So, the student scored equal to or more than 60 marks in only two subjects.

Answer: (B)

Let Manish's score be x ,

So Ajay's score = $x + 10$,

Rahul's score = $x - 5$

$$\text{Suresh's score} = 63 \times 3 - (x + x - 5)$$

$$= 194 - 2x$$

$$3(x + 10) = x - 5 + x + 194 - 2x + 90$$

$$3x + 30 = 279$$

$$3x = 249$$

$$x = 83$$

Sum of Manish and Suresh's score

$$= x + 194 - 2x = 111$$

15. **Answer: (C)**

Let father's present age in case the father did not die be X years

Then 3 years ago, it would have been $(X - 3)$ years & 12 years ago, it would have been $(X - 12)$ years

At the time of death of Arun's father, average age of the family of 6 members was 26 years.

So, total age of the family at that time 26×6
= 156 years

12 years ago, that is 9 years before the death
of his father, total age of these 6 members
will get reduced by $9 \times 6 = 54$

So, total age of the family excluding his
father 12 years ago from now = $156 - 54 =$
102

Average age of the family 12 years ago =
(Age of father + total age of other
members)/7 = 28

So, $((x - 12) + 102)/7 = 28$, on solving this
we get, $X = 106$ years

This would have been Arun's father present
age. He died three years ago, so at the time
of his death, Arun's father's age would have
been $106 - 3 = 103$ years

Hence, option c.

16. **Answer: (C)**

Total weight of A, B and C is $(65 \times 3) = 195$
kg

Total weight of A, B, C and D is $(68 \times 4) =$
272 kg

Weight of D = $(272 - 195) = 77$ kg

Weight of E = $77 - 4 = 73$ kg

Total weight of B, C, D and E = $(67 \times 4) =$
268 kg

Total weight of B, C and D
= $268 - 73 = 195$ kg

Weight of A = $(272 - 195) = 77$ kg.

17. **Answer: (A)**

Let the least amount of wages be Rs. x

$x + (x + 20) + (x + 40) + (x + 60) = 4 \times 60$

Or, $4x + 120 = 240$

Or, $4x = 240 - 120$

Or, $4x = 120$

So, $x = \text{Rs. } 30$.

18. **Answer: (C)**

Average age of all 40 teachers in 2007 = 35

Average age of all 40 teachers in 2009 = 37

Total age of 40 teachers in 2009 = 37×40
= 1480

Total age of 10 teachers who left in 2009
= 500

Total age of 30 teachers in 2009
= $1480 - 500 = 980$

Total age of 30 teachers in 2012
= $980 + 30 \times 3 = 1070$

Total age of 20 new teachers in 2012
= $20 \times 30 = 600$

Total age of 50 teachers in 2012
= $1070 + 600 = 1670$

Total age of 50 teachers in 2015
= $1670 + 50 \times 3 = 1820$

19. **Answer: (B)**

Now total age of 100 bankers in 2004 = 100
 $\times 50 + 2 \times 100 = 5200$.

Now 20 bankers whose average age was 60
retired.

So, total age now = $5200 - 1200 = 4000$.

Now total age of 80 bankers in 2007 = 4000
 $+ 3 \times 80 = 4240$.

Now, 40 bankers joined PNB in 2007 whose
average age was 38 years.

Hence, total age now = $4240 + 40 \times 38$
= 5760.

Now total age of 120 bankers in 2010
= $5760 + 3 \times 120 = 6120$.

Hence, average age in 2010 = $6120/120$
= 51 years.

20. **Answer: (C)**

Actual total marks of 100 girls
= $100 \times 92 - 40 = 9160$

Actual total marks of 80 boys
= $80 \times 124 + 20 = 9940$

Average marks of the class
= $(9160 + 9940)/180 = 106.1$